CE543: Introduction to Geotechnical Earthquake Design

Homework Assignment #5

February 20th, 2026

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Problem 1.

Consider the isotropically consolidated drained triaxial compression tests for specimens with an initial relative density of 100% by Lee and Seed (1967) in Figure 9 of the Monograph (Idriss and Boulanger, 2008; copied below).

(a) Plot the peak effective friction angle versus effective confining stress (σ_{3c}'). Above what effective confining stress does the peak effective friction angle become approximately constant for these specimens (i.e., equal to the constant volume friction angle)?

(b) On one figure, plot the Mohr circles at peak deviator stress for effective confining stresses of about 100, 290, and 1000 kPa. Draw a Mohr Coulomb failure surface to fit these test results (not as a straight line, but as an appropriately curved surface).

(c) Compute the final void ratio for both specimens (DR=100% and 38%) and all confining effectives stresses (σ_{3c}') on Figure 9, assuming that the reported volumetric strains are uniform throughout the specimens. Assume emin = 0.61 and emax = 1.03 for this sand. On one figure, plot the final void ratio versus initial confining stress. Do the computed void ratios converge to the same "critical state" void ratio at large strains? If not, explain why.

(d) Estimate the peak deviatoric stress that can be carried by a specimen with a relative density of 100% tested in ICD triaxial compression with an effective confining stress of 500 kPa (you can interpolate the data in the top left plot on Figure 9).

(e) If a specimen at DR=100% with an effective consolidation stress of 300 kPa was tested in undrained compression, would the pore water pressure be expected to increase or decrease? Would this sample be initially dense or loose of critical state? Explain your reasoning.

Hints:

Part (a): The peak effective friction angle for this sand should plateau at about 33 degrees at large effective stresses (larger than ~40 atm)

Part (c): For example, at confining stress of 116 atm, the loose sample (Dr=38%) has an initial void ratio of eo=0.87, final volumetric strain of ε_{vol}=-15.7%, and final void ratio of ef=0.576.

Part (d): $q_{peak} = \Delta\sigma_{d,peak}$ = 1750 kPa

Part (e): dense of critical expected to dilate.

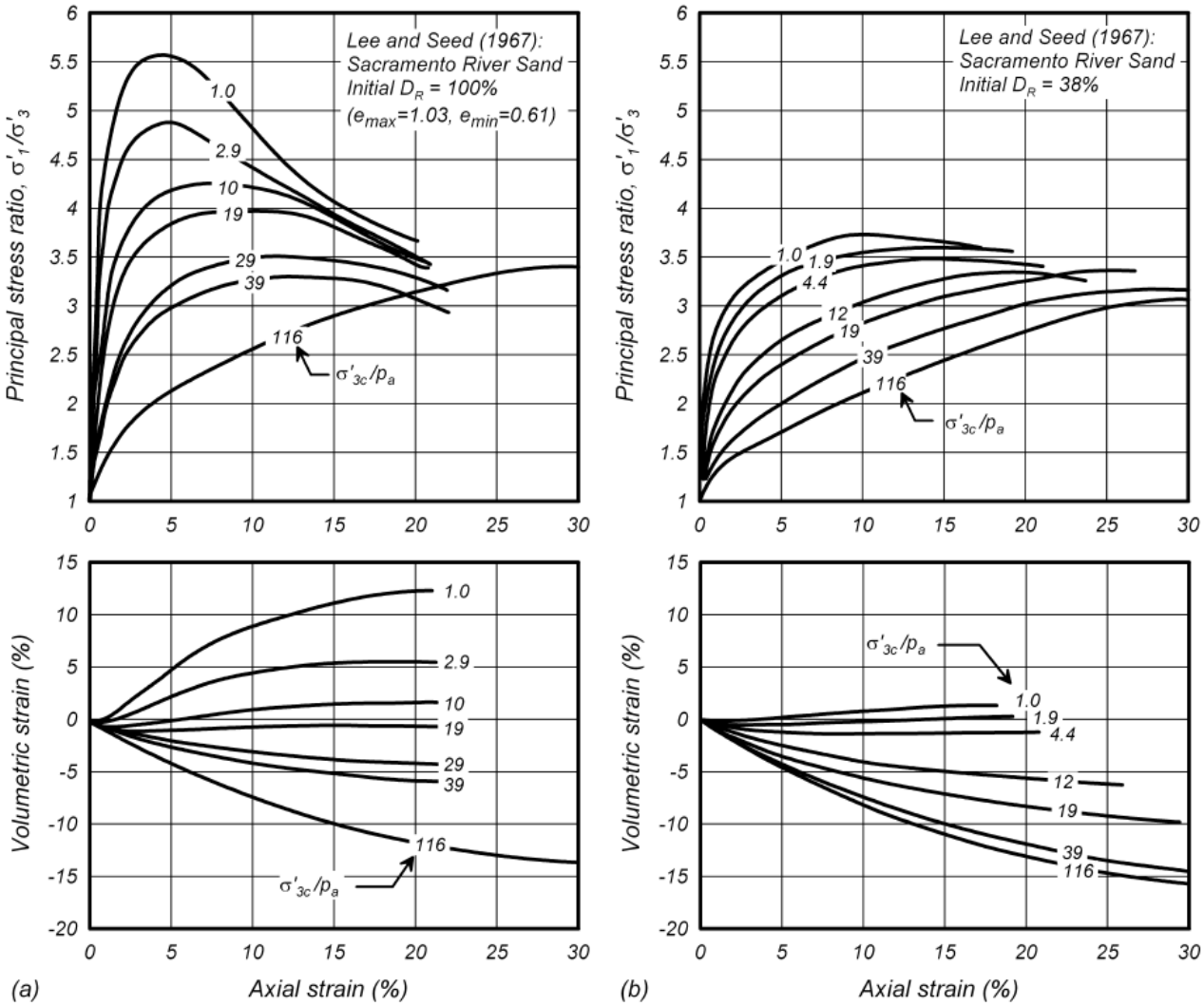

Figure 9. Monotonic loading response of dense ($D_R = 100\%$) and loose ($D_R = 38\%$) specimens of Sacramento River sand in drained triaxial compression tests, as shown in graphs (a) and (b) respectively (after Lee and Seed 1967, with permission from ASCE).

GIVEN:

Figure 9 from Lee & Seed (1967), considering the D_R = 100% sample.

FIND:

PART A

Plot the peak effective friction angle versus effective confining stress (σ3c'). Above what effective confining stress does the peak effective friction angle become approximately constant for these specimens (i.e., equal to the constant volume friction angle)?

PART B

On one figure, plot the Mohr circles at peak deviator stress for effective confining stresses of about 100, 290, and 1000 kPa. Draw a Mohr Coulomb failure surface to fit these test results (not as a straight line, but as an appropriately curved surface)?

PART C

Compute the final void ratio for both specimens (DR=100% and 38%) and all confining effective stresses (σ3c') on Figure 9, assuming that the reported volumetric strains are uniform throughout the specimens. Assume emin = 0.61 and emax = 1.03 for this sand. On one figure, plot the final void ratio versus initial confining stress. Do the computed void ratios converge to the same "critical state" void ratio at large strains? If not, explain why?

PART D

Estimate the peak deviatoric stress that can be carried by a specimen with a relative density of 100% tested in ICD triaxial compression with an effective confining stress of 500 kPa (you can interpolate the data in the top left plot on Figure 9).

PART E

If a specimen at DR=100% with an effective consolidation stress of 300 kPa was tested in undrained compression, would the pore water pressure be expected to increase or decrease? Would this sample be initially dense or loose of critical state? Explain your reasoning.

SOLUTION:

PART A

The peak effective friction angle, ϕ'_p, is related to the maximum flow number, N_{max} using the following relationship.

$$N_{max} = \frac{1 + sin(\phi'_p)}{1 - sin(\phi'_p)}$$

From the given figure, the maximum flow numbers per each series of confining stress can be collected. These values of N_{max} can then be tabulated with their values of ϕ'_p.

σ'_{3c} [atm]	N_{max}	$\phi'_p[\circ]$
1	5.6	44.2
2.9	4.9	41.4
10	4.3	38.5
19	4.0	36.9
29	3.6	34.4
39	3.3	32.3
116	3.4	33.1

Table S1. Values of cell pressure, max flow number, and peak effective friction angle.

A figure can then be generated of ϕ'_p with respect to σ'_{3c}, as seen in **Figure R1** where the constant volume friction angle is where the relationship between the two variables begins to 'flatten'.

PART B

The value of σ'_{3c} was found from *Part A* and can be multiplied by 100 kPa/atm to convert all values into the kPa space. The value of the major principal stress at peak is found using the following relationship.

$$\sigma'_{1p} = N_{max}\sigma'_{3c}$$

The peak deviatoric stress can then be found by subtracting the minor principal stress from the major principal stress.

$$q_p = \sigma'_{1p} - \sigma'_{3c}$$

Mohr's circles can then be generated for the listed confining pressures using the following relationships.

$$\text{radius at peak} \equiv r_f = \frac{q_p}{2}$$

$$\text{center at peak} \equiv c_f = \frac{\sigma'_{1p} + \sigma'_{3c}}{2}$$

$$\text{Mohr's circle stresses} \equiv (\sigma', \tau) = (c_p + r_p \cos\theta,\ r_p \sin\theta)\ \forall\ \theta\ \epsilon\ [0, 360]^\circ$$

With no cohesion, a quadratic line can be fit through the points of failure on the Mohr's circle stresses to obtain an approximated Mohr-Coulomb Failure Criterion.

$$\text{Normal Stress on Failure Criterion} \equiv \sigma'_{ff} = c_p - r_p \cos(90^\circ - \phi'_p)$$

$$\text{Shear Stress on Failure Criterion} \equiv \tau_{ff} = \sigma'_{ff} \tan(\phi'_p)$$

σ'_{3c} [atm]	σ'_{ff}[kPa]	τ_{ff}[kPa]
1	169.7	164.9
2.9	481.7	424.3
10	1622.6	1291.1

Table S2. Value on the Mohr-Coulomb Failure Criterion used for quadratic regression.

Using the tabulated values in **Table S2** and the origin, a quadratic regression can be used to make an estimate for a curved Mohr-Coulomb Failure Criterion. This regression assumes that stress units are in kPa.

$$\text{Mohr-Coulomb Failure Criterion Regression}: \tau_{ff} = -7.536\text{E-}05\sigma'^2_{ff} + 9.152\text{E-}01\sigma'_{ff} + 4.332$$

The Mohr's circles, the failure points along those circles, and the Mohr-Coulomb Failure Criterion can be plotted together as seen in **Figure S2.**

Part C

The initial void ratio, e_0, can be obtained using the following relationship.

$$D_R = \frac{e_{max} - e_0}{e_{max} - e_{min}}$$

Using the known values of e_{min} and e_{max} as given by the problem statement, the intial void ratios for the loose (i.e. D_R = 38% = 0.38) and dense (i.e. D_R = 100% = 1) are found to be 0.87 and 0.61 respectively.

The change in volumetric strain, $\Delta\varepsilon_{vol}$, can be obtained from the bottom graphs of Figure 9 for both loose and dense samples. $\Delta\varepsilon_{vol}$ or $\Delta\varepsilon_v$ is related to the change in void ratio, Δe in the following relationship.

$$\Delta\varepsilon_v = \frac{\Delta e}{1 + e_0} \Rightarrow \Delta e = \varepsilon_v(1 + e_0)$$

The final void ratio, e_f can then be obtained from the following relationship.

$$e_f = e_0 + \Delta e = e_0 + \varepsilon_v(1 + e_0)$$

The values are then tabulated for the loose and dense values, as seen in **Tables S2 and S3.** Once obtained, the values of final void ratio can be plotted with respect to σ'_{3c}.

$\varepsilon_v^{\text{loose}}$	σ'_{3c}[kPa]	e_f
0.015317	100.000000	0.899048721447930
0.003909	190.000000	0.877712282990977
-0.010473	440.000000	0.850810700186376
-0.060856	1200.000000	0.756574642770871
-0.096847	1900.000000	0.689257577370649
-0.143289	3900.000000	0.602391419143376
-0.157000	11600.000000	0.576747199999999

Table S2. Loose sample volumetric strains and final void ratios.

$\varepsilon_v^{\text{dense}}$	σ'_{3c}[kPa]	e_f
0.123720	100.000000	0.809189419795222
0.055631	290.000000	0.699566552901024
0.016809	1000.000000	0.637062286689420
-0.006485	1900.000000	0.599559726962457
-0.042321	2900.000000	0.541863481228669
-0.058447	3900.000000	0.515900170648464
-0.136092	11600.000000	0.390891638225256

Table S3. Dense sample volumetric strains and final void ratios.

Part D

The answer to this question is formatted in the *Results* section.

Part E

The answer to this question is formatted in the *Results* section.

Results:

Part A

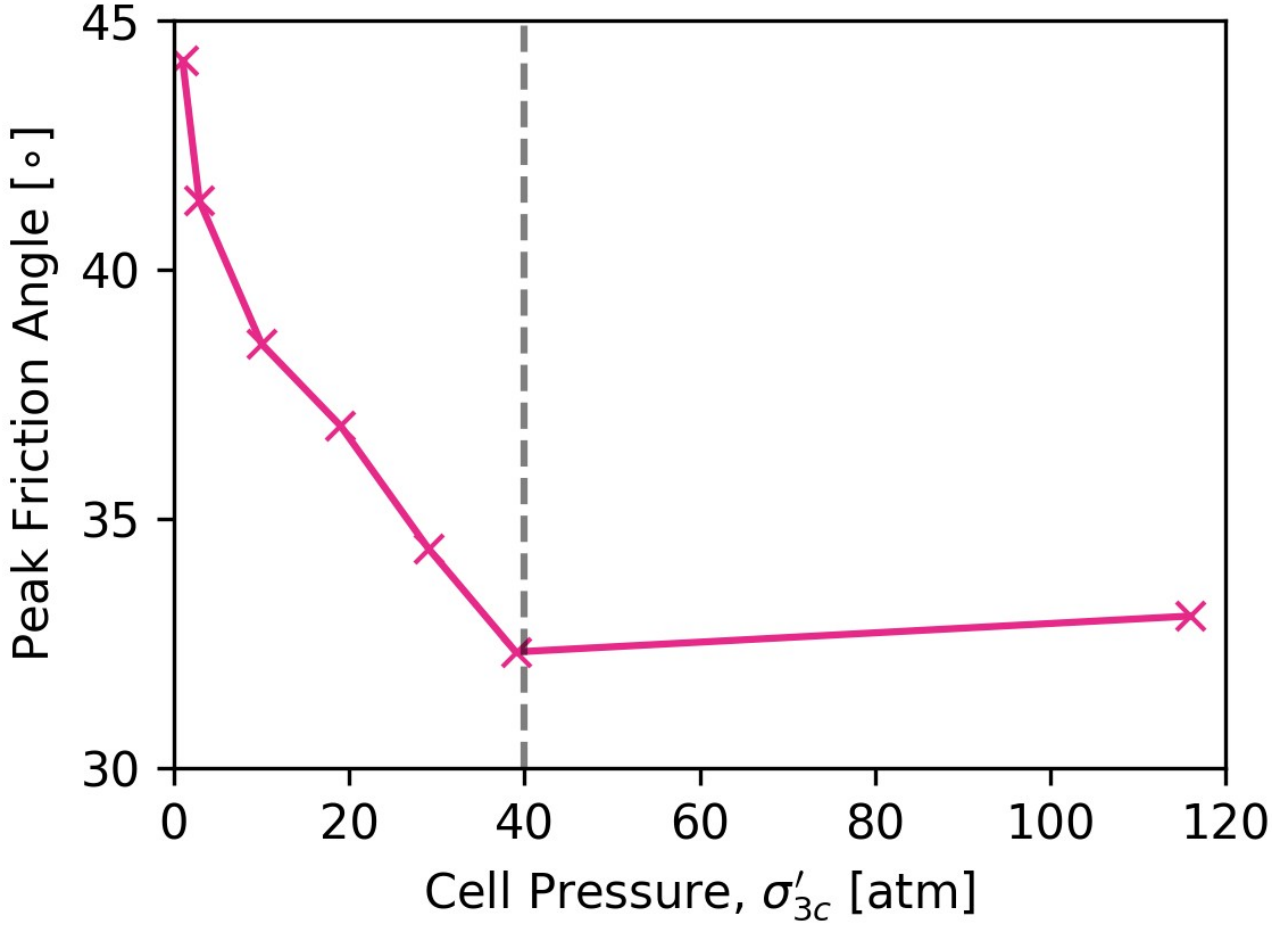

Figure R1. Peak effective friction angle plotted with respect to cell pressure.

The constant volume friction angle is around 33 degrees at a confining pressure of 40 atm.

PART B

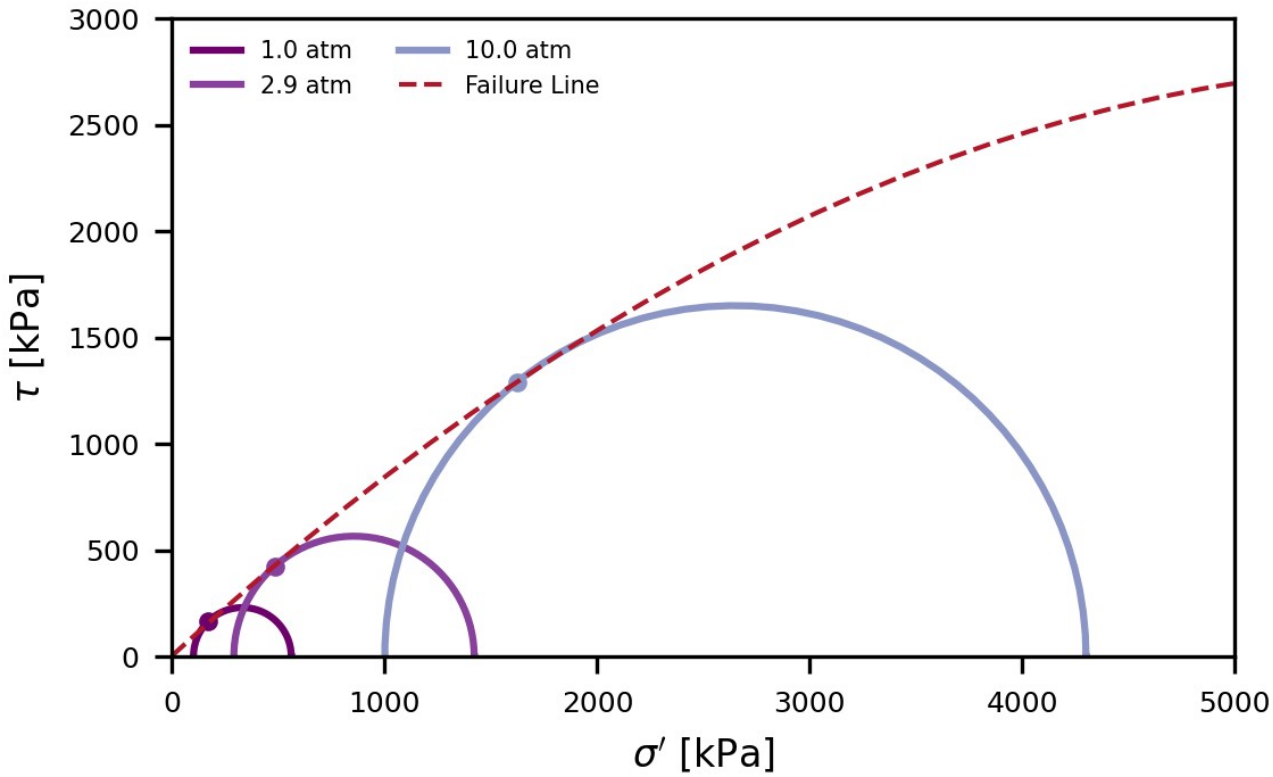

Figure R2. Mohr's circles at peak deviatoric stress with curved Mohr-Coulomb Failure Criterion line.

PART C

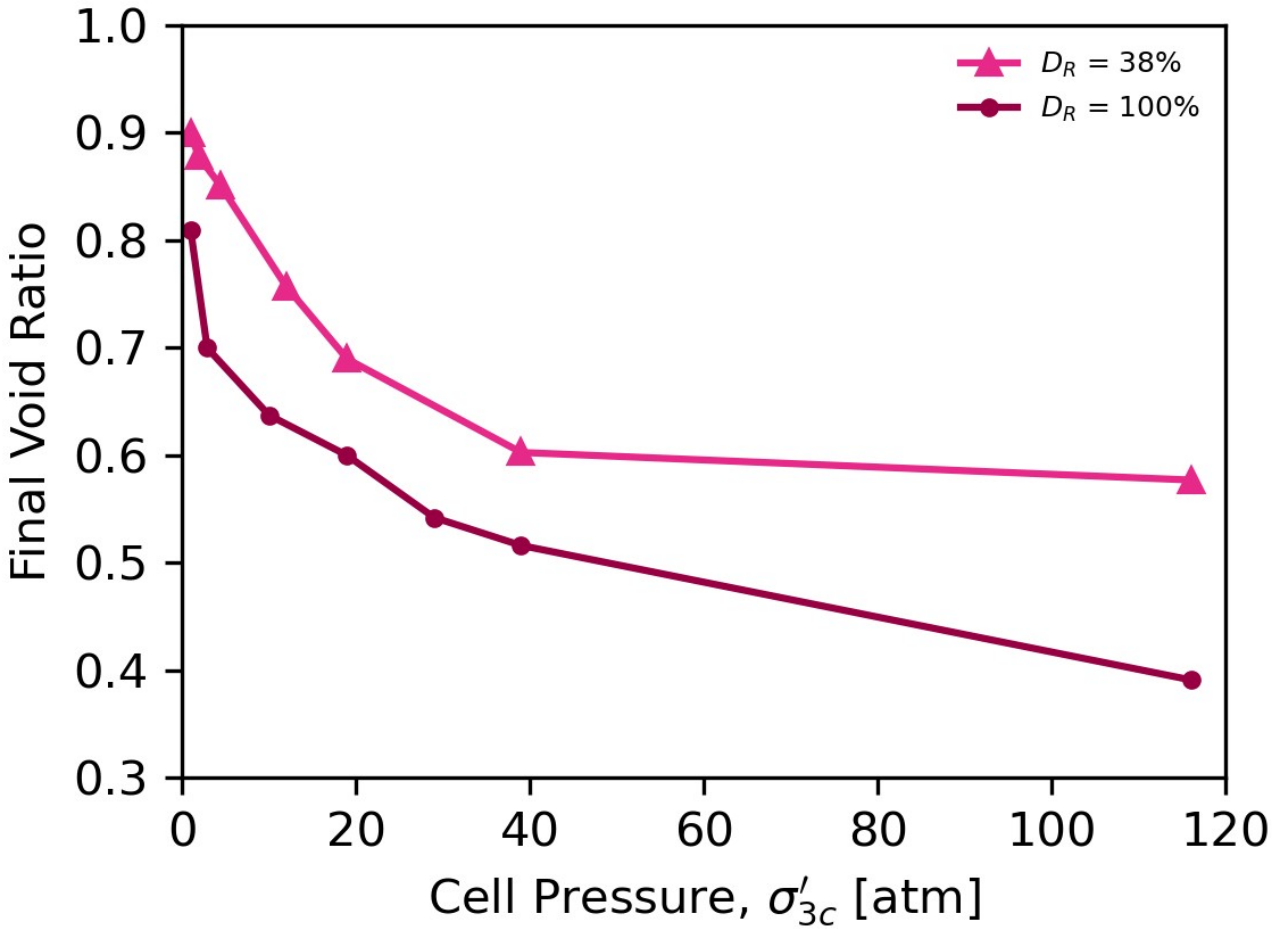

Figure R3. Final void ratio plotted with respect to cell pressure for loose and dense samples.

The samples do not appear to converge to the 'critical state' of the soil. This is likely due to limitations within the lab regarding the measurement of local void ratio within the shear band.

Part D

Interpolating the graph in *Figure 9*, a value of 500 kPa (i.e. 5 atm) confining stress provides a value of around 4.5 for N_{max}. This confining stress and flow number can be multiplied to obtain a value of peak major principal stress equal to 2250 kPa. Lastly, subtracting the peak major principal stress from the confining stress computes a peak deviatoric stress equal to appropriately 1750 kPa.

Part E

In a TX ICD in D_R = 100%, σ'_{3c} = 300 kPa, then according to the bottom left graph, the soil will have a positive $\Delta\varepsilon_{vol}$ which implies dilative tendencies. If a TX ICU is preformed on this sample with these initial conditions, the dilative behavior of this soil would result in negative pore water pressure, due to suction from attempts of the sample to expand.

Problem 2.

Consider the undrained triaxial compression test result for specimen B of Toyoura sand in Figure 12 (Ishihara 1993) of the Monograph:

(a) What is the constant volume (or critical state) effective friction angle, φ_{cv}'?

(b) On a single figure, plot the Mohr circles for the specimen at the following times: (i) the end of consolidation, (ii) at phase transformation (i.e., the transition from incrementally contractive to incrementally dilative behavior), and (iii) at critical state.

(c) Determine the excess pore pressure (Δu) at point Q.

Hints:

Part (a): 30.9 degrees

Part (c): 71 kPa

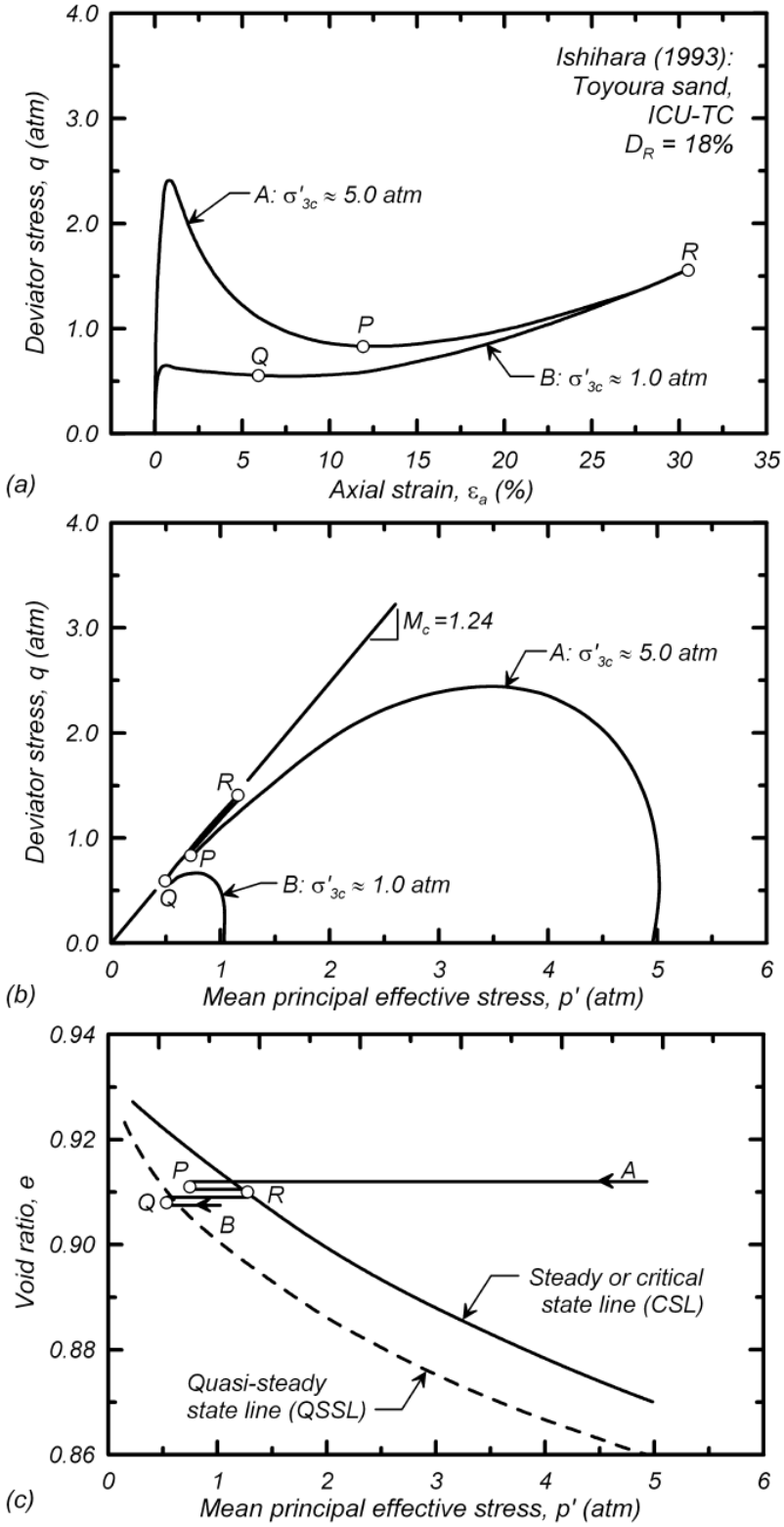

Figure 12. Quasi-steady-state (QSS) and steady-state behavior of very loose Toyoura sand in ICU triaxial compression tests (after Ishihara 1993).

GIVEN:

Sample B in Figure 12 from Ishihara (1993).

FIND:

PART A

Determine the critical state friction angle, ϕ'_{cs}.

PART B

Plot the Mohr's circles for the consolidation phase, phase transformation phase (located at Point Q), and the critical state.

PART C

Determine the excess pore water pressure, Δu, at Point Q.

SOLUTION:

PART A

The slope of the M-line is given as 1.24 from the center graph. The following relationship can be used to solve for the critical state friction angle.

$$M = \frac{3\sin\phi'_{cs}}{6 - \sin\phi'_{cs}}$$

PART B

During consolidation phase, there is no deviatoric stress. The effective confining stress for Sample B is given as 1 atm (100 kPa). This means that at the end of consolidation, the Mohr's circle will be a single point located at the value of the effective stress equal to the cell pressure.

Phase transformation is stated to be located at Point Q. At this location, the deviatoric stress is 0.5 atm (50 kPa) and thus the major principal stress is equal to the sum of the deviatoric stress and the confining stress.

The deviatoric stress at critical state is equal to 160 kPa. Once again, the sum of the deviatoric stress and the confining stress is equal to the major principal stress at this location.

The following equations can be used to obtain the radius and center of the Mohr's circles.

$$\text{radius} \equiv r = \frac{q}{2}$$

$$\text{center} \equiv c = \frac{\sigma'_{1p} + \sigma'_{3c}}{2}$$

PART C

The peak mean effective stress at Point Q is found to be just below 0.5 atm (i.e. p' = 49 kPa). The deviatoric stress at Point Q is equal to 0.6 atm (600 kPa). The cell pressure used to consolidate Sample B is 100 kPa. The below relationship can be solved to for the excess pore water pressure.

$$p' = \sigma'_{3c} + \frac{1}{3}q - \Delta u$$

Results:

Part A

The effective critical state friction angle is equal to 30.9 degrees.

Part B

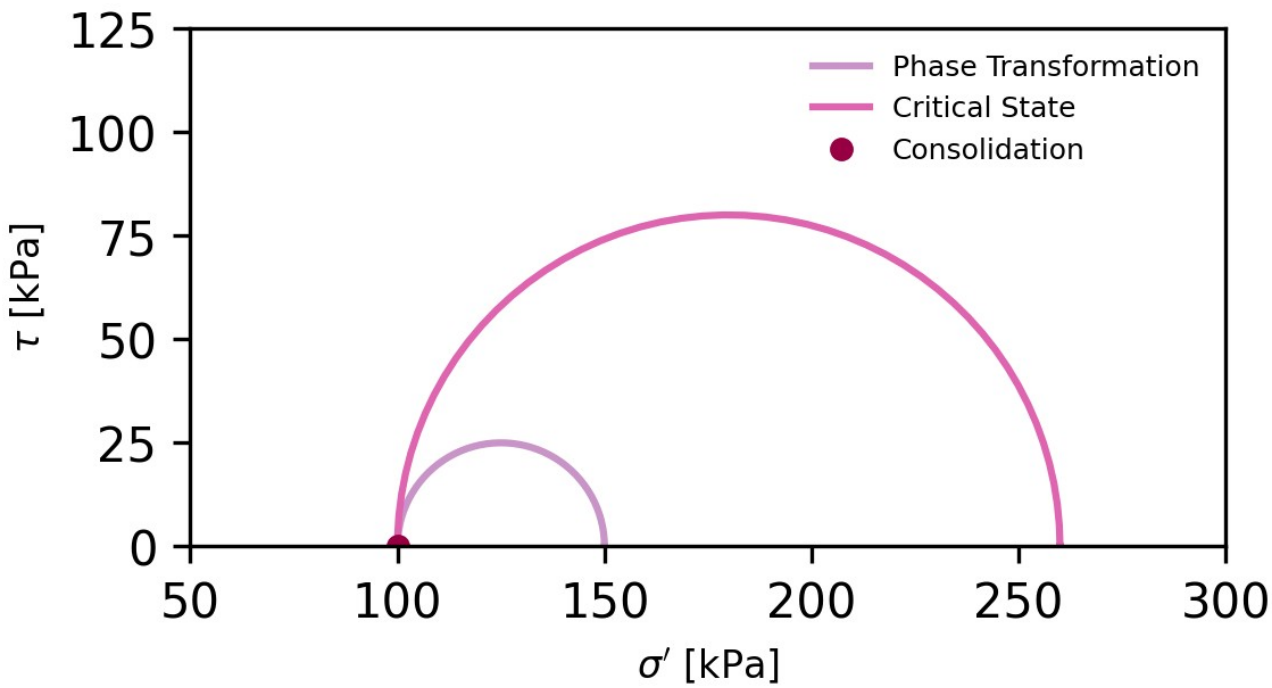

Figure R4. Mohr's circles at specified states.

State	σ_3'[kPa]	σ_3'[kPa]	Radius [kPa]	Center [kPa]
Consolidation	100	100	0	100
Phase Transformation	100	150	50	125
Critical State	100	260	80	180

Table R1. Values for the Mohr's circles at specified states.

Part C

The excess pore water pressure is equal to 71 kPa.

Problem 3.

Consider the undrained triaxial compression test data for Toyoura sand summarized in Figure 12 of the Monograph.

(a) For a specimen that was isotropically consolidated at an effective confining stress of 200 kPa and an initial void ratio of 0.920, and then tested in undrained compression, estimate the critical state values for the following quantities: σ_3', σ_1', Δu, ru, p' and q.

(b) For a specimen that was isotropically consolidated at an effective confining stress of 200 kPa and an initial void ratio of 0.880, and then tested in undrained compression, estimate the critical state values for the following quantities: σ_3', σ_1', Δu, ru, p', and q.

(c) If the specimens in (a) and (b) had been loaded in drained triaxial compression, would they have been contractive, dilative, or not changed volume at all? Would they be stronger in drained or undrained loading?

Hints:

Part (a): Δu = 164 kPa

Part (b): Δu = -29 kPa

Part (c): Sample B is dense of critical, dilative, negative PWP in undrained, stronger in undrained compared to drained.

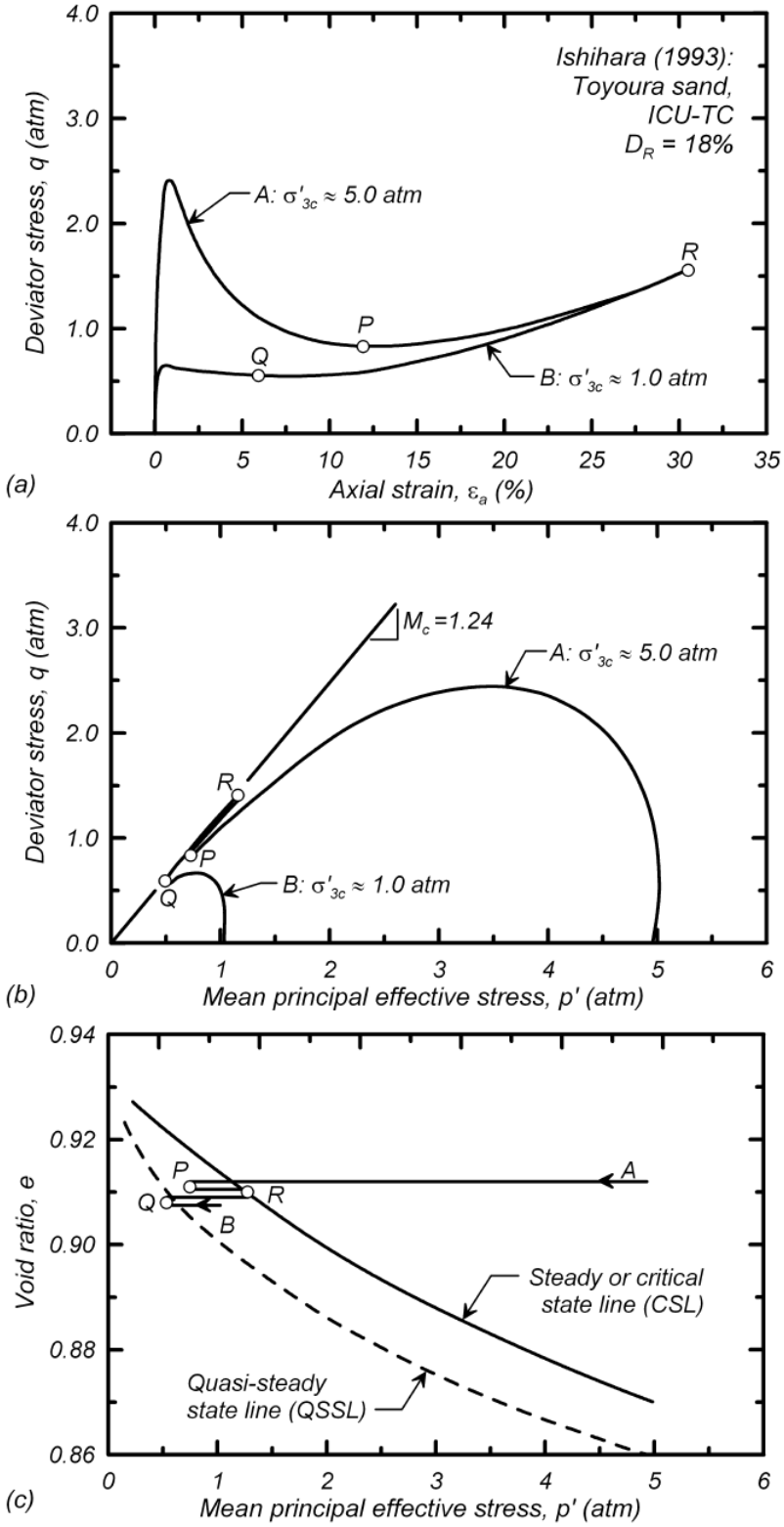

Figure 12. Quasi-steady-state (QSS) and steady-state behavior of very loose Toyoura sand in ICU triaxial compression tests (after Ishihara 1993).

GIVEN:

Figure 12 from Ishihara (1993).

FIND:

PART A

For a specimen that was isotropically consolidated at an effective confining stress of 200 kPa and an initial void ratio of 0.920, and then tested in undrained compression, estimate the critical state values for the following quantities: σ_3', σ_1', Δu, r_u, p' and q.

PART B

For a specimen that was isotropically consolidated at an effective confining stress of 200 kPa and an initial void ratio of 0.880, and then tested in undrained compression, estimate the critical state values for the following quantities: σ3', σ1', Δu, r_u, p', and q.

PART C

If the specimens in *Part A* and *Part B* had been loaded in drained triaxial compression, would they have been contractive, dilative, or not changed volume at all? Would they be stronger in drained or undrained loading?

SOLUTION:

PART A

Using the bottom graph from *Figure 12* from Ishihara (1993), the (x,y) is equal to (p' [atm], e). The point (2 atm, 0.920) is located to the right of the CSL where the end of consolidation is. In a TX ICU, the sample will have to move left to get to the CSL, since the volume is constant. If the volume were not constant, the sample would be able to move downwards to the CS, along the void ratio axis. This means this sample would exhibit contractive tendencies. In Critical State Space, the value at failure will be equal to (0.60 atm, 0.920). The void ratio remains constant in a TX ICU test, since volume is constant.

In the Stress Path Plot, the slope of the M-line can be used to obtain the exact value of the deviatoric stress at failure. This can be done using the following relationship.

$$q_{cs} = M \times p'[\text{atm}]$$

Once q_{cs} is obtained, it can be used to determine the value of σ'_{3cs} also known as σ_3'. σ'_{3cs} is equal to the cell pressure minus any developed pore water pressure during failure. In equation form, the following then must be true.

$$\sigma'_3 = \sigma'_{3c} - \Delta u$$

$$p' = \sigma'_{3c} + \frac{1}{3}q_{cs} - \Delta u = \sigma'_3 - \frac{1}{3}q_{cs}$$

Once σ_3' is found, it is simple to calculate Δu. σ_{1cs}' or σ_1' is found via the following relationship.

$$\sigma_1' = \sigma_3' + q_{cs}$$

Finally, the value of the pressure ratio, r_u, is found using the following relationship.

$$r_u = \frac{\Delta u}{\sigma_{3c}'}$$

Part B

Using the bottom graph from *Figure 12* from Ishihara (1993), the (x,y) is equal to (p' [atm], e). The point (2 atm, 0.880) is located to the left of the CSL where the end of consolidation is. In a TX ICU, the sample will have to move right to get to the CSL, since the volume is constant. If the volume were not constant, the sample would be able to move upwards to the CSL, along the void ratio axis. This means this sample would exhibit dilative tendencies. In Critical State Space, the value at failure will be equal to (3.75 atm, 0.888).

A similar line of analysis to *Part* A may be followed for this sample.

Part C

The answers to this question will be formatted in the *Results* section.

RESULTS:

PART A, PART B

e_0	σ'_{3c}[kPa]	p' [kPa]	q_{cs}[kPa]	σ'_3[kPa]	Δu[kPa]	σ'_1[kPa]	r_u
0.920	200	60	74	35	165	109	0.825
0.880	200	355	477	226	-26	703	-0.130

Table R2. Answers to *Part A* and *Part B*.

PART C

The sample from *Part A* exhibits contractive tendencies and is thus also considered loose of critical. The generation of positive pore water pressure decreases the effective stress on the sample, which therefore decreases the intergranular resistance due to friction. This sample would be stronger in a drained test, where the positive excess pore water pressure cannot develop.

The sample from *Part B* exhibits dilative tendencies and is thus also considered dense of critical. The generation of negative pore water pressure increases the effective stress on the sample, which therefore increase the intergranular resistance due to friction. This sample would be weaker in a drained test, where the negative excess pore water pressure cannot develop.